Skills Summary

Solving Systems Algebraically by Substitution or Elimination

Use this reference while completing your worksheet or when studying.

A solution is one pair (x, y) that makes *both* equations true. Solving algebraically means getting rid of one variable so a single equation in the other one is left.

- **Choose the method from what you see.** A variable already alone (or one step from it) means substitute. Matching or opposite coefficients mean add or subtract. Otherwise scale first.
- Whatever you do to one side of an equation, do to the whole side: scaling $3x + 2y = 16$ by 2 gives $6x + 4y = 32$, and the solutions do not change.
- Find one variable, substitute back for the other, then check the pair in **both** original equations. The check is not optional — it catches every sign slip.

1. Substitution

Use it when one equation says $y = \ldots$ or $x = \ldots$, or becomes that in one step. Replace that variable in the *other* equation — never in the one you took it from, which collapses to $0 = 0$ and tells you nothing.

Example: $x + y = 21$ and $y = 2x$.
Step 1. The second equation already gives y by itself, so substituting costs no rearranging at all.
Step 2. Replacing y in the first equation: $x + 2x = 21$, so $3x = 21$ and $x = 7$; then $y = 2(7)$.

$$\boldsymbol{x = 7,\ y = 14}$$

Try it: $x + y = 19$ and $x - y = 5$. Rearrange the second to $x = y + 5$, then substitute.

check: $\boldsymbol{x = 12,\ y = 7}$

2. Elimination by adding or subtracting

Line the equations up. If a variable's coefficients are **opposites**, add. If they are **identical**, subtract. Nothing needs multiplying, and one variable disappears in a single step.

Example: $2x + y = 11$ and $3x - y = 9$.
Step 1. The y terms are $+y$ and $-y$, which are opposites, so adding the two equations is the one move that removes a variable outright.
Step 2. Adding: $5x = 20$, so $x = 4$; then $2(4) + y = 11$ gives $y = 3$.

$$\boldsymbol{x = 4,\ y = 3}$$

Try it: $x + 4y = 18$ and $x + 2y = 12$. The x terms are identical — subtract.

check: $\boldsymbol{x = 6,\ y = 3}$

3. Elimination after scaling

When no coefficient matches, multiply one or both equations until one variable's coefficients match or are opposites, then add or subtract. Pick the variable with the smaller common multiple — it is less arithmetic and fewer places to slip.

Example: $2x + 3y = 12$ and $3x - y = 7$.
Step 1. No coefficients match, so scale for the cheapest common multiple: the y terms 3 and 1 need only the second equation tripled.
Step 2. $3(3x - y = 7)$ gives $9x - 3y = 21$; adding the first equation gives $11x = 33$, so $x = 3$, and $3(3) - y = 7$ gives $y = 2$.

$$\boldsymbol{x = 3,\ y = 2}$$

Try it: $3x + 2y = 19$ and $2x + 5y = 31$. Scale both — for example by 2 and 3 — to eliminate x.

check: $\boldsymbol{x = 3,\ y = 5}$

When the variables all vanish. If elimination leaves a false statement such as $0 = 4$, the lines are parallel and there is **no solution**. If it leaves a true statement such as $0 = 0$, the two equations describe the same line and there are **infinitely many** solutions. Neither is a mistake — each is the answer, and the sentence you write is what earns the mark.